Bioinf pipelines HW1

Install nextflow

```
~/pers/u/biopipelines ❯ nextflow -version                                  base at 21:52:44

      N E X T F L O W
      version 25.10.4 build 11173
      created 10-02-2026 15:17 UTC (18:17 MSK)
      cite doi:10.1038/nbt.3820
      http://nextflow.io

~/pers/u/biopipelines ❯                                                    base at 21:55:00
```

Run the pipeline on the test data in:

1. Docker

```
~/pers/u/biopipelines ❯ nextflow run nextflow-io/rnaseq-nf -profile docker

N E X T F L O W  ~  version 25.10.4

Launching `https://github.com/nextflow-io/rnaseq-nf` [happy_picasso] DSL2 - revision: ca20a6dfd2 [master
]

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /Users/nechn/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_1_48850000_49020000.Gg
al71.500bpflank.fa
reads        : /Users/nechn/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_gut_{1,2}.fq
outdir       : results

executor >  local (4)
[5e/896b65] RNASEQ:INDEX (ggal_1_48850000_49020000) | 1 of 1 ✔
[4d/e411fa] RNASEQ:FASTQC (ggal_gut)                | 1 of 1 ✔
[43/1cddbf] RNASEQ:QUANT (ggal_gut)                 | 1 of 1 ✔
[f7/e28b16] MULTIQC                                 | 1 of 1 ✔

Done! Open the following report in your browser --> results/multiqc_report.html

Completed at: 09-Apr-2026 21:50:58
Duration    : 2m 4s
CPU hours   : 0.1
Succeeded   : 4


~/pers/u/biopipelines ❯ ls                                  took ⌛ 2m 7s  base at 21:50:58
drwxr-xr-x@   - nechn  9 Apr 21:50 ▪ .nextflow
.rw-r--r--@ 13k nechn  9 Apr 21:50 ▤ .nextflow.log
.rw-r--r--@ 17k nechn  9 Apr 21:44 ▪ .nextflow.log.1
.rw-r--r--@ 16k nechn  9 Apr 21:42 ▪ .nextflow.log.2
drwxr-xr-x@   - nechn  9 Apr 21:50 ▪ results
drwxr-xr-x@   - nechn  9 Apr 21:50 ▪ work
~/pers/u/biopipelines ❯                                                    base at 21:52:44
```

2. Conda environment

```
~/pers/u/biopipelines ❯ nextflow run . -profile conda                    base at 21:55:23

N E X T F L O W  ~  version 25.10.4

Missing project main script: ./main.nf
~/pers/u/biopipelines ❯ git clone https://github.com/nextflow-io/rnaseq-nf    base at 21:55:26
Cloning into 'rnaseq-nf'...
remote: Enumerating objects: 1036, done.
remote: Counting objects: 100% (640/640), done.
remote: Compressing objects: 100% (242/242), done.
remote: Total 1036 (delta 419), reused 539 (delta 346), pack-reused 396 (from 2)
Receiving objects: 100% (1036/1036), 666.07 KiB | 2.47 MiB/s, done.
Resolving deltas: 100% (506/506), done.
~/pers/u/biopipelines ❯ cd rnaseq-nf                                     base at 21:55:46
~/pers/u/biopi/rnaseq-nf on master ❯ nextflow run . -profile conda       base at 21:55:50

N E X T F L O W  ~  version 25.10.4

Launching `./main.nf` [modest_jepsen] DSL2 - revision: 295b10c8d2

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /Users/nechn/personal/university/biopipelines/rnaseq-nf/data/ggal/ggal_1_48850000_49020000.Ggal71.500bpflank.fa
reads        : /Users/nechn/personal/university/biopipelines/rnaseq-nf/data/ggal/ggal_gut_{1,2}.fq
outdir       : results

[-        ] RNASEQ:INDEX  -
[-        ] RNASEQ:FASTQC -
executor >  local (1)
[-        ] RNASEQ:INDEX          -
executor >  local (1)
[-        ] RNASEQ:INDEX          -
executor >  local (1)
[-        ] RNASEQ:INDEX          -
executor >  local (3)
executor >  local (4)
[83/d49e56] RNASEQ:INDEX (ggal_1_48850000_49020000) | 1 of 1 ✔
[2d/2e93e6] RNASEQ:FASTQC (ggal_gut)                | 1 of 1 ✔
[e1/908bb2] RNASEQ:QUANT (ggal_gut)                 | 1 of 1 ✔
[aa/90bee6] MULTIQC                                 | 1 of 1 ✔

Done! Open the following report in your browser --> results/multiqc_report.html

Completed at: 09-Apr-2026 21:57:51
Duration    : 1m 57s
CPU hours   : (a few seconds)
Succeeded   : 4

~/pers/u/biopi/rnaseq-nf on master ❯                  took ⌛ 1m 59s  base at 21:57:51
```

Download any RNA-seq dataset from NCBI (but not to big) and run the pipeline with any profile on it
https://www.ncbi.nlm.nih.gov/sra/SRX32838157[accn]

$ prefetch SRR37982863
$ fastq-dump --split-files --gzip SRR37982863
$ curl -O https://ftp.ensembl.org/pub/release-113/fasta/mus_musculus/cdna/Mus_musculus.GRCm39.cdna.all.fa.gz

Then it took forever to run the pipeline on the real 3.5 Gb pair reads data so I took only 1k reads and reran.

$ gunzip -c SRR37982863_1.fastq.gz | head -n 4000 | gzip > SRR37982863_sub_1.fastq.gz
$ gunzip -c SRR37982863_2.fastq.gz | head -n 4000 | gzip > SRR37982863_sub_2.fastq.gz

```
~/pers/u/biopipelines ❯ nextflow run rnaseq-nf/main.nf \
    -profile docker \
    --reads "$PWD/mousebrain/SRR37982863_sub_{1,2}.fastq.gz" \
    --transcriptome "$PWD/mousebrain/Mus_musculus.GRCm39.cdna.all.fa.gz" \
    --outdir results_mouse_sub

N E X T F L O W  ~  version 25.10.4

Launching `rnaseq-nf/main.nf` [astonishing_lavoisier] DSL2 - revision: 295b10c8d2

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /Users/nechn/personal/university/biopipelines/mousebrain/Mus_musculus.GRCm39.cdna.all.fa.
gz
reads        : /Users/nechn/personal/university/biopipelines/mousebrain/SRR37982863_sub_{1,2}.fastq.gz
outdir       : results_mouse_sub

executor >  local (4)
[87/c65b1b] RNASEQ:INDEX (Mus_musculus)      | 1 of 1 ✔
[89/e5b72e] RNASEQ:FASTQC (SRR37982863_sub)  | 1 of 1 ✔
[96/8d681e] RNASEQ:QUANT (SRR37982863_sub)   | 1 of 1 ✔
[d5/57ef6d] MULTIQC                          | 1 of 1 ✔

Done! Open the following report in your browser --> results_mouse_sub/multiqc_report.html

Completed at: 09-Apr-2026 23:31:37
Duration    : 4m 22s
CPU hours   : 0.1
Succeeded   : 4


~/pers/u/biopipelines ❯                                   took ⌛ 4m 24s ◆ base at ◷ 23:31:37
```